

Cointegrated Currency Linkages During Crises

Abstract

This paper examines the long-run cointegration and short-run dynamics among major foreign exchange (FX) markets across multiple global crises, focusing on the Global Financial Crisis (GFC, 2007–2009) and the COVID-19 pandemic (2020–2022). Using daily spot rates for five major currency pairs (EUR/USD, USD/JPY, GBP/USD, USD/CHF, AUD/USD) from January 2000 to December 2025, we apply Johansen cointegration tests, vector error correction models (VECMs), generalized variance decomposition, and network-based measures to identify regime-dependent shifts in currency market integration.

Our results confirm that during the GFC, cointegration among major currencies strengthened, consistent with theory predicting reduced cross-currency autonomy under USD funding stress and heightened global risk aversion. In contrast, the COVID-19 period exhibits a surprising weakening of long-run linkages—despite elevated volatility—suggesting increased currency-specific autonomy, possibly driven by heterogeneous policy responses, capital controls, and reserve management strategies. This divergence is statistically significant in Johansen trace statistics across alternative COVID sample windows and robust to lag selection.

The findings contribute new evidence that global crises need not uniformly increase FX integration: while the GFC compressed long-run currency autonomy, COVID-19 partially broke long-run parity. This challenges the standard crisis-contagion narrative and underscores the importance of incorporating regime-dependent pass-through parameters into structural models of exchange rates. Our results have implications for international portfolio diversification, risk management, and the understanding of safe-haven behavior under heterogeneous crisis conditions.

1 Introduction

Foreign exchange (FX) markets are the largest and most liquid financial markets in the world, underpinning global trade, investment flows, and the international transmission of monetary

policy. The degree of integration among currency markets has far-reaching implications for portfolio diversification, hedging strategies, and the propagation of financial shocks across borders. Episodes of systemic stress—most notably the Global Financial Crisis (GFC) of 2007–2009 and the COVID-19 pandemic of 2020–2022—offer natural laboratories to examine how extreme events reshape both the short-run dynamics and long-run equilibria of exchange rates.

Motivation. A central question in international finance is whether major currencies share a common long-run stochastic trend driven by global forces, or whether certain currencies retain “autonomy” from global shocks. Asset-pricing theory suggests that in frictionless, fully integrated markets with a single global stochastic discount factor (SDF), bilateral exchange rates vis-à-vis the U.S. dollar (USD) should be cointegrated, with a rank of $N - 1$ in an N -currency system (Backus, Foresi, and Telmer, 2001; Engel and West, 2005). In practice, structural frictions, heterogeneous monetary regimes, and episodic market stress can cause deviations from this benchmark. The GFC was characterized by heightened USD funding stress (Baba and Packer, 2009; McGuire and von Peter, 2009), which theory predicts should reduce cross-currency autonomy and tighten long-run linkages. In contrast, our empirical results for the COVID-19 period indicate an unexpected weakening of cointegration, suggesting greater currency-specific independence than standard crisis-contagion narratives would imply.

Novelty and Contribution. While the building blocks of our framework—log-normal SDFs, factor-driven exchange rate dynamics (Verdelhan, 2010), and Johansen cointegration analysis (Johansen and Juselius, 1990)—are well established, our contribution is to synthesize these elements into a unified, *regime-dependent* model that is structurally interpretable and empirically testable. Specifically, we introduce a tractable “autonomy” parameter $\psi_{i,r}$ that governs the pass-through of a single global USD factor into currency i during regime r . The parameter allows for systematic variation in sensitivity across market conditions: crises such as the GFC compress $\psi_{i,r}$ toward zero, reflecting diminished autonomy, whereas certain post-crisis periods show recovery in autonomy, particularly for commodity-linked and reserve-rich economies (Aizenman and Jinjara, 2014). The COVID-19 episode, however, emerges as a novel case where autonomy *increases* in spite of global stress, challenging the conventional crisis integration paradigm.

Empirical Integration. The model provides a direct theoretical mapping to our empirical toolkit. The single-common-trend structure implies a cointegration rank of $N - 1$ under

full integration; deviations from this rank reveal changes in market structure. The $\psi_{i,r}$ parameters have a clear econometric counterpart in the adjustment coefficients and long-run correlation patterns estimated from Vector Error Correction Models (VECMs). Elevated global factor volatility $\sigma_{g,r}^2$ and reduced autonomy map into higher long-run correlations and stronger network connectivity among currencies (Flood and Rose, 1995; Inoue, 2012). Conversely, increased $\psi_{i,r}$ manifests as weaker cointegration and greater dispersion in currency linkages—precisely what we observe during the COVID-19 period.

Position in the Literature. Our work bridges structural asset-pricing theory and empirical cointegration analysis in a way not previously applied to joint major-and-emerging-market FX systems across multiple global crises. Existing studies tend to focus on short-run volatility spillovers (Melvin and Taylor, 2009; Aloui et al., 2014) or on regional blocs (Kanas, 2005), whereas we integrate long-run equilibrium analysis, regime-switching pass-through, and network-based diagnostics. We also contribute to the literature on crisis-induced structural change in FX markets by documenting and structurally interpreting the apparent breakdown of long-run parity during COVID-19—a phenomenon not observed in the GFC and, to our knowledge, not previously highlighted in the cointegration literature.

Roadmap. Section ?? describes our dataset of daily Tuesday-close spot rates for major currencies from January 2000 to December 2025, with sub-samples for the GFC, post-GFC, and COVID periods. Section ?? details our unit root testing, Johansen cointegration framework, and the theoretical mapping from the autonomy parameter to econometric measures. Section ?? presents the empirical findings, contrasting the GFC and COVID regimes. Section ?? discusses theoretical and policy implications, and Section ?? concludes.

2 Literature Review

Research on FX market integration often focuses on correlations and volatility spillovers. Early studies such as (author?) (4) identified common stochastic trends in major exchange rates, while more recent work has used cointegration to explore long-run parity conditions (16). During crises, the literature finds elevated correlations and reduced diversification benefits (? ?). The GFC brought renewed attention to currency linkages through the lens of USD funding markets (6; 3; 18). For emerging markets, (author?) (7) emphasize the role of capital flows and policy responses in shaping post-crisis exchange rate behavior.

Literature Gap

Despite a growing body of research on financial market contagion during the GFC, the long-run integration of currency markets across distinct regimes—pre-crisis, crisis, and post-crisis—remains underexplored. Existing studies primarily focus on short-term volatility spillovers or regional clusters of currencies (e.g., ASEAN, Eurozone) and rarely integrate both major and emerging FX markets into a unified framework. Moreover, most analyses emphasize temporary contagion during the crisis without testing whether structural relationships in currency markets persisted or reversed in the post-crisis era.

This paper addresses these gaps by providing a comprehensive examination of currency market linkages using a broad panel of major and emerging FX pairs, deploying Johansen cointegration, variance decomposition, and network-based measures to capture both long-run equilibrium relationships and dynamic transmission channels across regimes.

3 Literature Review

Long-run comovement and cointegration in FX

A core strand of the FX literature documents common stochastic trends among major exchange rates and uses cointegration methods to study long-run parity relations. Early evidence shows that several dollar exchange rates share persistent common drivers, motivating vector error-correction representations and rank tests (Baillie and Bollerslev, 1989; Johansen, 1988, 1991). In parallel, asset-pricing perspectives explain why exchange rates can look close to random walks at short horizons even when fundamentals matter in present-value terms (Engel and West, 2005). These insights underlie the widespread use of Johansen’s framework to test whether N dollar exchange rates admit $N - 1$ cointegrating relations in integrated markets.

Risk-based and intermediary-based views of FX

Modern finance models connect exchange rates to priced risks and intermediary constraints. Currency returns load on common factors (“dollar” and “carry”), which help explain the cross-section of risk premia (Lustig, Roussanov and Verdelhan, 2011). Intermediary asset-pricing adds that exchange rates adjust with dealers’ balance-sheet capacity; in these models, shifts in international liquidity and limits to arbitrage can reprice currencies and alter comovement (Gabaix and Maggiori, 2015). These perspectives provide structural channels—risk prices and funding constraints—through which “global” shocks can synchronize

currencies while allowing for time-varying heterogeneity across countries.

Crises, funding stress, and the global dollar system

Crisis episodes reveal how dollar funding conditions transmit globally. During the GFC, USD funding markets were impaired, with sharp deviations in FX swaps, heightened margins, and a measurable shortage of dollars outside the U.S. banking system (Baba and Packer, 2009; McGuire and von Peter, 2009). Deviations from covered interest parity (CIP) crystallized the role of balance-sheet constraints; the post-crisis literature formalized these deviations as equilibrium objects linked to dealer leverage and regulations (Du, Tepper and Verdelhan, 2018). These mechanisms suggest that, in stress, currencies may load more strongly and more similarly on the global dollar factor, potentially tightening long-run linkages even if short-run dynamics are turbulent.

COVID-19 as a distinct stress event

The COVID-19 shock produced an exceptional “dash for dollars,” dislocations in the FX swap market, and unprecedented policy responses (central bank swap lines and facilities). BIS evidence documents severe March 2020 stress in FX swaps and a subsequent normalization as swap lines expanded (Schrimpf, Shin and Sushko, 2020). Related work shows that swap lines alleviated dollar funding strains and reduced CIP bases across jurisdictions (Bahaj and Reis, 2020, among others). While these papers emphasize funding markets and short-run dislocations, they open the door to testing whether COVID also altered *long-run* linkages—potentially in ways that differ from the GFC.

Emerging markets, resilience, and policy

Beyond majors, emerging market (EM) currencies are shaped by external finance, commodity terms of trade, and policy regimes. Post-GFC analyses report that EMs’ resilience depended on pre-existing fundamentals, macroprudential tools, and reserve buffers (Didier, Hevia and Schmukler, 2012; Aizenman and Jinjark, 2014). These features can modulate pass-through from global factors, implying that the strength of long-run comovement is regime- and country-dependent.

Structural breaks, regime dependence, and inference

From a time-series perspective, detecting regime changes in cointegration is crucial. Residual-based tests allow for single endogenously dated breaks in cointegrating vectors (Gregory and

Hansen, 1996), and multiple-break frameworks offer inference on shifts in high-dimensional systems (Qu and Perron, 2007). Given the GFC and COVID chronology, these tools are natural complements to Johansen tests for assessing whether cointegration weakens, disappears, or re-emerges across regimes.

Connectedness and network approaches

A complementary literature measures the *strength* and *direction* of cross-market spillovers using variance-decomposition-based connectedness (Diebold and Yilmaz, 2012) and graph-theoretic tools applied to FX data (e.g., minimum-spanning trees/PMFGs). These methods help separate the number of common trends (a rank concept) from how tightly the system moves together (a covariance/connectedness concept), which is especially informative during crises.

How this paper advances the literature

This paper contributes along four fronts:

1. **Unified theory—empirics for regimes.** We embed a no-arbitrage, log-normal SDF with nonstationary global factors into a pass-through system that *explicitly maps* to Johansen cointegration and VECM objects. The framework nests both the classic $N - 1$ cointegration benchmark (single global trend) and multi-trend regimes where rank declines—an element often missing in studies focused solely on correlations or short-run spillovers (cf. Johansen, 1991; Du, Tepper and Verdelhan, 2018).
2. **Autonomy as structural heterogeneity.** We introduce currency- and regime-specific autonomy parameters that govern factor pass-through, providing a structural interpretation for cross-country differences emphasized in EM policy work (Aizenman and Jinjark, 2014) and for time-varying comovement seen during the GFC and COVID (Baba and Packer, 2009; Schrimpf, Shin and Sushko, 2020).
3. **Crisis comparison on *rank* vs. *tightness*.** We distinguish the *number* of common trends (cointegration rank) from the *tightness* of co-movement (long-run covariance/connectedness). This clarifies how a crisis can feature high correlations yet *lower* cointegration if new nonstationary drivers become active (a possibility raised by intermediary-based models and documented funding dislocations).
4. **COVID as a structural outlier.** Whereas the GFC fits the narrative of compressed autonomy and tight linkages, we show that COVID can coincide with weaker cointe-

gration among majors—consistent with distinct funding mechanisms, policy backstops (swap lines), and sectoral shocks—thereby updating the crisis-integration paradigm for modern dollar funding architecture (Schrumpf, Shin and Sushko, 2020; Bahaj and Reis, 2020).

Taken together, the paper connects classic cointegration evidence (Baillie and Bollerslev, 1989; Johansen, 1988, 1991) to contemporary finance mechanisms—risk factors, intermediary constraints, and policy backstops—providing a structural interpretation for why long-run FX linkages can strengthen in one crisis (GFC) yet weaken in another (COVID)

Comparison to Existing Studies

Model Setup

We now present a regime-dependent factor model linking the long-run dynamics of N non-USD exchange rates to a set of nonstationary global factors. The U.S. is the numeraire. Let the spot exchange rate be

$$S_{i,t} = \text{units of currency } i \text{ per USD} \quad (i = 1, \dots, N),$$

so an increase in $S_{i,t}$ indicates USD appreciation against currency i . Define

$$X_{i,t} = \log S_{i,t}, \quad X_t = (X_{1,t}, \dots, X_{N,t})'.$$

2. Pricing Kernel (SDF)

We price assets in USD using a log-normal stochastic discount factor (SDF):

$$\log M_{t+1} = -r_t - \frac{1}{2} \varsigma_r^2 - \xi_{t+1}, \tag{1}$$

where r_t is the USD short rate, ξ_{t+1} is the SDF innovation, and $\mathbb{E}_t[M_{t+1}] = e^{-r_t}$ by construction.

Multi-factor global structure. In regime r , the innovation admits q_r nonstationary global factors:

$$\xi_{t+1} = \lambda'_r g_{t+1} + v_{t+1}, \quad \mathbb{E}_t[g_{t+1}v_{t+1}] = 0, \tag{2}$$

where:

Table 1: Comparison of Existing Studies on FX Linkages and This Paper

Study	Publication Outlet	Scope & Methodology	How This Paper Improves
Zhang & Li (2016)	<i>International Review of Economics and Finance</i>	ASEAN currencies; Johansen cointegration; pre/during/post GFC.	Extends to global panel of major + EM FX; uses variance decomposition and network analysis beyond ASEAN.
Inagaki (2013)	<i>Journal of International Financial Markets, Institutions and Money</i>	Major FX spillovers; GARCH-based volatility contagion.	Goes beyond short-run volatility; examines long-run cointegration and structural regime shifts.
Hwang & Min (2012)	<i>Economic Modelling</i>	Asian currency integration; VECM pre/post GFC.	Incorporates emerging + major FX jointly; explicitly tests crisis-period structural breaks and post-crisis decoupling.
Habib & Stracca (2010)	<i>Journal of International Money and Finance</i>	Safe-haven behavior of USD, JPY, CHF in crisis.	Extends to broader FX network; analyzes co-movement dynamics and diversification implications across regimes.
Aloui, Ben Aïssa & Nguyen (2014)	<i>Emerging Markets Review</i>	EM currency contagion during GFC; correlation-based.	Uses Johansen, variance decomposition, and network metrics; compares pre/during/post GFC to show regime-dependent integration.

- $g_t = (g_{1,t}, \dots, g_{q_r,t})'$ are q_r independent $I(1)$ global factors, e.g., USD funding stress, global risk appetite, commodity-price shocks.
- $\lambda_r \in \mathbb{R}^{q_r}$ are SDF-level factor loadings in regime r .
- $v_{t+1} \sim \mathcal{N}(0, \sigma_v^2)$ is an idiosyncratic USD pricing-kernel shock orthogonal to g_{t+1} .

Each factor follows a random walk with regime-dependent drift and volatility:

$$g_{k,t+1} = g_{k,t} + \mu_{k,r} + \sigma_{g_{k,r}} \varepsilon_{k,t+1}, \quad \varepsilon_{k,t+1} \stackrel{i.i.d.}{\sim} \mathcal{N}(0, 1), \quad k = 1, \dots, q_r. \quad (3)$$

The total SDF innovation variance is:

$$\zeta_r^2 = \lambda_r' \Sigma_{g,r} \lambda_r + \sigma_v^2, \quad \Sigma_{g,r} = \text{diag}(\sigma_{g_{1,r}}^2, \dots, \sigma_{g_{q_r,r}}^2).$$

The currencies each have varying exposures to the vector of global factors (e.g., global risk, liquidity conditions, macroeconomic policy shifts), which means that the common-factor block loads differently on each currency: for some currencies, one or more factor loadings are large (high coefficients along specific factor directions), while for others they are small or negligible. In (??), the innovation to the stochastic discount factor is decomposed into two conceptually distinct components: a *systematic* term, $\lambda_r^\top \mathbf{g}_{t+1}$, and an *idiosyncratic* term, v_{t+1} . The q_r -dimensional vector $\mathbf{g}_{t+1}$ captures shocks that are common—though not necessarily equally important—to all currencies; examples include shifts in USD funding conditions, coordinated changes in major central bank policy stances, or global risk-on/risk-off episodes. The vector λ_r measures the sensitivity of the *pricing kernel* to each of these common shocks in regime r , and in the affine solution each currency i inherits its own q_r -vector of pass-through coefficients $\lambda_{i,r}$, reflecting its specific exposure profile to the different global factors. By contrast, v_{t+1} represents purely currency-specific noise in the SDF, orthogonal to $\mathbf{g}_{t+1}$ by construction. Its variance σ_v^2 fully determines its scale, so no additional factor loadings are needed at the SDF level. Economically, the vector λ_r captures how each orthogonal component of the common-factor block propagates through the pricing kernel, while the heterogeneous $\lambda_{i,r}$ map those common shocks into individual currency returns; the idiosyncratic term v_{t+1} is already asset-specific and its effects are entirely absorbed by the currency it directly influences.

3. Payoffs, No-Arbitrage and Affine Solution

A one-period claim pays $D_{i,t+1}$ units of currency i ; its USD price is

$$P_{i,t} = \mathbb{E}_t \left[M_{t+1} \frac{D_{i,t+1}}{S_{i,t+1}} \right]. \quad (4)$$

For a pure FX claim delivering one unit of currency i (i.e. $D_{i,t+1} = 1$), replication implies

$P_{i,t} = 1/S_{i,t}$. No-arbitrage gives:

$$\frac{1}{S_{i,t}} = \mathbb{E}_t \left[M_{t+1} \frac{1}{S_{i,t+1}} \right],$$

or equivalently,

$$1 = \mathbb{E}_t \left[M_{t+1} \frac{S_{i,t}}{S_{i,t+1}} \right] = \mathbb{E}_t \left[M_{t+1} e^{X_{i,t} - X_{i,t+1}} \right], \quad (5)$$

our key FX pricing restriction.

The intuition behind the no-(riskless)-arbitrage condition is that the dollar price paid today for a future claim of one unit of foreign currency

$$\mathbb{E}_t \left[M_{t+1} \frac{1}{S_{i,t+1}} \right],$$

must equal the current dollar cost of acquiring that same unit of foreign currency,

$$\frac{1}{S_{i,t}},$$

If this equality does not hold, an investor could exploit the difference to earn a profit without risk. Any profit that involves bearing risk is not considered a riskless arbitrage profit and is therefore outside the scope of our model.

4. Solving for Exchange-Rate Levels

Assuming (i) the SDF is log-normal as in (??)–(??), and (ii) fundamentals z_t ($K \times 1$) are stationary and independent of the global-factor innovations, the log-linear solution takes the affine form

$$X_{i,t} = \mu_{i,r} + \lambda'_{i,r} g_t + \Gamma'_i z_t + u_{i,t}, \quad (6)$$

with $u_{i,t}$ stationary, and

$$\mu_{i,r} = -\bar{r}_r - (\lambda_{i,r} + \lambda_r)^\top \mu_r + \kappa_{i,r}, \quad \text{with} \quad \kappa_{i,r} \equiv \frac{1}{2} [(\lambda_{i,r} + \lambda_r)^\top \Sigma_{g,r} (\lambda_{i,r} + \lambda_r) + \sigma_v^2 + \Delta(\Gamma'_i z_t + u_{i,t})] \quad (7)$$

$$\Gamma'_i z_t = \sum_{k=1}^K \gamma_{i,k} z_{k,t}, \quad z_t = (z_{1,t}, \dots, z_{K,t})' \text{ is stationary}, \quad (8)$$

$$u_{i,t} = X_{i,t} - \mu_{i,r} - \lambda'_{i,r} g_t - \Gamma'_i z_t, \quad u_{i,t} \text{ stationary}. \quad (9)$$

Here $\bar{r}_r \equiv \mathbb{E}[r_t \mid r]$ is the regime- r mean short rate; $(\mu_r, \Sigma_{g,r}, \lambda_r)$ are regime parameters of the q_r -dimensional global-factor process and SDF; and σ_v^2 is the variance of the SDF idiosyncratic residual.

The global-factor vector g_t enters linearly in the pricing kernel, so $X_{i,t}$ must load linearly on g_t . Currencies differ in their exposure profiles to the q_r orthogonal global factors, with some exhibiting large loadings in certain factor directions and others having smaller or near-zero loadings. This heterogeneity is captured by the currency-specific vector $\lambda_{i,r}$, while λ_r measures the sensitivity of the pricing kernel to each global factor in regime r . In the SDF, the innovation decomposition

$$\xi_{t+1} = \lambda_r^\top g_{t+1} + v_{t+1}$$

separates the *systematic* component, $\lambda_r^\top g_{t+1}$, from the *idiosyncratic* component, v_{t+1} . The vector g_{t+1} represents shocks common to all currencies—such as shifts in global risk appetite, changes in major reserve currency funding conditions, or synchronized monetary policy moves—but whose economic impact varies across currencies according to their $\lambda_{i,r}$ profiles. By contrast, v_{t+1} is orthogonal to g_{t+1} by construction, representing purely currency-specific noise in the SDF; its variance σ_v^2 fully determines its scale, so no additional factor loadings are needed for this term at the SDF level.

The vector z_t is taken to be stationary so that exchange rates can load on *mean-reverting* conditions without introducing additional stochastic trends. While short-horizon exchange rates often approximate random walks—weakening their level relations with trending fundamentals¹—stationary variables capture time-varying but transitory forces that help explain cross-sectional heterogeneity and conditional dynamics.² Treating z_t as stationary preserves the common $I(1)$ trends driven solely by g_t and leaves the cointegration rank determined by the rank of Λ_r ; standard VECM/CVAR frameworks allow such stationary exogenous controls without affecting rank tests.³

Finally, the residual $u_{i,t}$ measures the deviation of $X_{i,t}$ from the level implied by its long-run relation with g_t and z_t . If $u_{i,t}$ were nonstationary, it would imply an additional persistent factor orthogonal to g_t , contradicting the assumption that g_t spans all nonstationary variation. Stationarity of $u_{i,t}$ ensures that any cointegration between $X_{i,t}$ and g_t is not impaired, and confines $u_{i,t}$ to mean-reverting, thereby preserving the internal consistency of the model’s long-run factor structure.

¹See Engel and West (2005) for the near-random-walk implication; classic out-of-sample evidence includes Meese and Rogoff (1983).

²Taylor-rule fundamentals and related stationary deviations have documented predictive content at certain horizons; see, e.g., Molodtsova and Papell (2009).

³See Johansen’s VECM framework as commonly implemented.

The USD price of one unit of currency i delivered at $t + 1$ is $1/S_{i,t}$. No-arbitrage implies:

$$1 = \mathbb{E}_t [M_{t+1} e^{X_{i,t} - X_{i,t+1}}].$$

Solving under log-normality and stationary fundamentals z_t yields:

$$X_{i,t} = \mu_{i,r} + \lambda'_{i,r} g_t + \Gamma'_i z_t + u_{i,t}, \quad (10)$$

where $\lambda_{i,r} \in \mathbb{R}^{q_r}$ reflects currency-specific exposure to the q_r global factors, $g_t \in \mathbb{R}^{q_r}$.

5. Regime-Dependent Pass-Through (“Autonomy”)

In order to model how the sensitivity of exchange rates to the global factor vector changes across regimes, we parameterize the factor loadings for currency i in regime r as

$$\lambda_{i,r} = (1 - \psi_{i,r}) \odot \bar{\lambda}_i, \quad \bar{\lambda}_i \neq 0 \text{ fixed}, \quad (11)$$

where $\psi_{i,r} = (\psi_{i1,r}, \dots, \psi_{iq_r,r})' \in [0, 1]^{q_r}$ is the vector of “autonomy” parameters of currency i from each global factor in regime r , $\bar{\lambda}_i \in \mathbb{R}^{q_r}$ is the baseline (regime-invariant) vector of factor loadings, and $\odot$ denotes the Hadamard (elementwise) product. A lower $\psi_{ik,r}$ implies stronger pass-through from factor k (larger $|\lambda_{ik,r}|$), meaning that the currency’s value is more sensitive to movements in that global factor. A higher $\psi_{ik,r}$ implies greater autonomy from factor k in that regime.

The following assumptions describe how volatility and autonomy evolve across regimes, and ensure the statistical tractability of the model:

- A1** For at least one factor k , $\sigma_{g_k, \text{crisis}}^2 > \sigma_{g_k, \text{pre}}^2$ and $\sigma_{g_k, \text{crisis}}^2 \geq \sigma_{g_k, \text{post}}^2$. Crisis periods are characterized by heightened volatility in one or more global factors, reflecting increased uncertainty, tighter global financial conditions, or both.
- A2** For all i, k , $\psi_{ik, \text{crisis}} \leq \psi_{ik, \text{pre}}$. Autonomy compresses during crises as market stress and funding pressures force more synchronized exchange-rate movements with the global factors.
- A3** For a non-empty set $\mathcal{E}$ (emerging markets) and for at least one factor k , $\psi_{ik, \text{post}} > \psi_{ik, \text{crisis}}$. Certain EMs regain autonomy from one or more global factors after crises due to policy adjustments, reserve accumulation, or reduced reliance on specific channels of external funding.

A4 z_t and $u_{i,t}$ are stationary and independent of the global-factor innovations ε_{t+1} . Stationarity of z_t ensures they capture mean-reverting fundamentals without introducing new stochastic trends; stationarity of $u_{i,t}$ preserves cointegration between $X_{i,t}$ and its drivers, while independence from ε_{t+1} maintains orthogonality between predictable components and shocks.

Factor-loading structure. For each regime r , define the $N \times q_r$ pass-through matrix

$$\Lambda_r = \begin{bmatrix} \lambda'_{1,r} \\ \vdots \\ \lambda'_{N,r} \end{bmatrix} = \begin{bmatrix} (1 - \psi_{11,r}) \bar{\lambda}_{11} & \dots & (1 - \psi_{1q_r,r}) \bar{\lambda}_{1q_r} \\ \vdots & \ddots & \vdots \\ (1 - \psi_{N1,r}) \bar{\lambda}_{N1} & \dots & (1 - \psi_{Nq_r,r}) \bar{\lambda}_{Nq_r} \end{bmatrix},$$

which stacks the currency-specific loading vectors $\lambda'_{i,r}$ across i .

Here:

- $\lambda_{i,r} \in \mathbb{R}^{q_r}$ are the currency-specific loadings on the q_r global factors in regime r .
- $\psi_{ik,r}$ is the “autonomy” of currency i from factor k in regime r ; lower values imply higher pass-through from that factor.
- Γ_i loads on stationary fundamentals z_t , which do not alter the cointegration rank.
- $u_{i,t}$ is stationary and mean-reverting.

Exchange-rate system. With these definitions, the cross-section of exchange rates in regime r is

$$X_t = \mu_r + \Lambda_r g_t + \Gamma z_t + u_t, \tag{12}$$

where $\mu_r \in \mathbb{R}^N$ collects the regime-specific intercepts, $g_t \in \mathbb{R}^{q_r}$ is the vector of nonstationary global factors, Γ is the $N \times K$ matrix of stationary-factor loadings, and u_t is the vector of stationary residuals.

4. Cointegration Rank and Regime Dependence

Given the affine representation

$$X_t = \mu_r + \Lambda_r g_t + \Gamma z_t + u_t,$$

the long-run dimension of comovement among the N exchange rates in regime r is determined by the rank of the pass-through matrix Λ_r .

The vector $g_t \in \mathbb{R}^{q_r}$ collects the q_r nonstationary global factors relevant in regime r , while $\Lambda_r \in \mathbb{R}^{N \times q_r}$ stacks the currency-specific loading vectors $\lambda'_{i,r}$. Stationary fundamentals z_t and residuals u_t do not affect the cointegration rank because they are $I(0)$.

The number of distinct common stochastic trends transmitted to X_t in regime r is

$$\text{rank}(\Lambda_r) \leq \min(N, q_r).$$

This yields the standard cointegration rank result:

- If $\text{rank}(\Lambda_r) = 1$, there is a *single* nonstationary common component, and the system has $N - 1$ cointegrating relations (the single-factor baseline).
- If $\text{rank}(\Lambda_r) = k > 1$, there are $N - k$ cointegrating relations (multiple independent global trends drive the system).
- If $\text{rank}(\Lambda_r) = N$, there is no cointegration (each currency contains its own stochastic trend).

Regime variation. Both the factor volatilities $\sigma_{gk,r}^2$ and the autonomy parameters $\psi_{ik,r}$ can vary across regimes, altering the effective rank of Λ_r . For example:

- A crisis may *increase* factor volatilities (making existing common trends more dominant) and/or *compress* autonomy parameters (aligning currencies more closely with the factors).
- If a new global factor emerges—e.g., a pandemic-specific risk driver during COVID—it can raise $\text{rank}(\Lambda_r)$ and reduce the number of cointegrating relations. In the extreme, if enough independent trends are introduced, cointegration may disappear even if it was present pre-crisis.

5. Key Regime Assumptions

We impose the following regime-dependent structure on volatilities and autonomy:

A1 Crisis volatilities: $\sigma_{gk,\text{crisis}}^2 \geq \sigma_{gk,\text{pre}}^2$ for all k , with strict inequality for at least one k . *Crisis periods exhibit heightened volatility in at least one global factor, reflecting increased uncertainty and tighter global financial conditions.*

A2 Autonomy compression: $\psi_{ik,\text{crisis}} \leq \psi_{ik,\text{pre}}$ for all i, k . *Market stress and funding pressures force more synchronized exchange-rate responses to the global factors.*

A3 Post-crisis partial recovery: For some emerging markets $i \in \mathcal{E}$ and some factors k , $\psi_{ik,\text{post}} > \psi_{ik,\text{crisis}}$. *These currencies regain some autonomy after crises through policy adjustments, reserve accumulation, or reduced reliance on specific funding channels.*

A4 Stationarity and orthogonality: z_t and u_t are stationary and orthogonal to factor innovations ε_{t+1} . *This preserves the cointegration rank implied by Λ_r and maintains orthogonality between predictable components and shocks.*

6. Main Propositions

The following propositions summarize the key theoretical implications of the multi-factor, regime-dependent pass-through model developed above. They connect the statistical properties of the pass-through matrix Λ_r to observable patterns of cointegration, long-run co-movement, and regime shifts in exchange-rate dynamics. Formal proofs are provided in the Appendix.

Proposition 1 (Rank–Cointegration Link). Given regime r and assumptions A1–A4, the cointegration rank of X_t is

$$\text{Cointegration rank} = N - \text{rank}(\Lambda_r).$$

Economic interpretation: This result formalizes the direct link between the number of independent nonstationary global trends and the strength of long-run exchange-rate linkages. If $\text{rank}(\Lambda_r) = 1$, all persistent variation is driven by a single global factor (e.g., the USD funding/liquidity factor), yielding $N - 1$ independent stationary spreads. As $\text{rank}(\Lambda_r)$ increases, additional nonstationary drivers—such as commodity-price trends, regional funding shocks, or segmented risk premia—introduce new sources of long-run divergence, reducing the number of cointegrating relations. In the extreme case $\text{rank}(\Lambda_r) = N$, each currency has its own independent trend and no long-run stationary spread survives.

Proposition 2 (Crisis Co-Movement Amplification). If $\text{rank}(\Lambda_r)$ remains unchanged in crisis, but (i) $\sigma_{gk,r}^2$ rises for one or more factors, and (ii) $|\lambda_{ik,r}|$ converge across currencies (autonomy compression), then long-run correlations increase.

Economic interpretation: Crises amplify co-movement through two reinforcing channels:

1. **Variance channel:** Higher volatility in the common factors increases their share of total long-horizon variance.

2. **Loading-alignment channel:** Lower dispersion in $\psi_{ik,r}$ makes currency responses more similar in magnitude and sign, reducing idiosyncratic offsets.

Even without introducing new stochastic trends, these channels produce tighter long-run alignment—a pattern consistent with the “risk-on/risk-off” behavior observed during global financial stress.

Proposition 3 (Rank Change in Crisis). If new persistent global factors emerge or autonomy shifts such that $\text{rank}(\Lambda_{\text{crisis}}) > \text{rank}(\Lambda_{\text{pre}})$, the number of cointegrating relations falls. If $\text{rank}(\Lambda_{\text{crisis}}) = N$, cointegration disappears entirely.

Economic interpretation: Certain crises—such as the COVID-19 shock—introduce structural changes that are not spanned by pre-crisis factors: e.g., pandemic-specific risk premia, supply-chain dislocations, or regime-specific commodity shocks. If these shocks load persistently across currencies but in a pattern independent of pre-existing factors, they increase the effective dimension of g_t and the rank of Λ_r . The result is fewer cointegrating relations, with the extreme of rank N corresponding to a complete breakdown of long-run linkage.

Proposition 4 (Post-Crisis Divergence). If $\text{rank}(\Lambda_{\text{post}}) < \text{rank}(\Lambda_{\text{crisis}})$ and the dispersion in autonomy parameters $\psi_{ik,r}$ widens, long-run correlations weaken.

Economic interpretation: The dissipation of certain crisis-specific global factors in the post-crisis period reduces $\text{rank}(\Lambda_r)$, potentially restoring some cointegrating relations. However, as countries pursue differentiated policy responses, alter reserve portfolios, or shift trade and financial linkages, heterogeneity in autonomy parameters may increase. This heterogeneity reduces the absolute correlation of long-horizon returns, producing a “partial decoupling” effect in which some pre-crisis integration is permanently lost.

4 Data

Our dataset consists of **daily spot exchange rates** recorded at the **Tuesday New York close**, which reduces the impact of weekend-related noise and overlapping Monday/Friday effects. The sample spans **January 2000 to December 2025**, covering both tranquil and crisis periods in the global foreign exchange market. All exchange rates are quoted **against the U.S. dollar (USD)**. Prices are transformed to natural logarithms for analysis, and returns are calculated as first differences of log prices.

Currencies. We focus on the most liquid “major” currency pairs, representing a blend of global reserve currencies, safe-haven assets, and commodity-linked units:

- **EUR/USD** — Euro, the second most important reserve currency and the deepest non-USD FX market.
- **GBP/USD** — British pound, historically significant but subject to geopolitical regime shifts (e.g., Brexit).
- **USD/JPY** — Japanese yen, a major safe-haven asset with persistent interest rate differentials.
- **USD/CHF** — Swiss franc, another canonical safe-haven currency.
- **AUD/USD** — Australian dollar, a commodity-linked currency influenced by terms-of-trade shocks.

Crisis Subperiods. To examine stability and regime dependence in long-run currency relationships, we split the sample into key subperiods:

1. **Global Financial Crisis (GFC):** August 2007 – June 2009.
2. **COVID-19 Crisis:** Core pandemic phase from February 2020 – December 2021, with extended windows (October 2019 – December 2022) to capture pre-crisis build-up and post-crisis adjustment.
3. **Post-COVID:** January 2022 – August 2025, enabling tests of whether long-run equilibrium linkages were restored.

These subperiods align with macro-financial chronology and allow direct comparison of cointegration stability across regimes.

5 Methodology

5.1 Unit Root Tests

We verify the order of integration of each exchange rate series using **Augmented Dickey–Fuller (ADF)** and **Phillips–Perron (PP)** tests:

- **Levels:** Tests include intercepts (and deterministic trends when appropriate) to confirm nonstationarity $I(1)$.

- **First Differences:** Confirm stationarity $I(0)$ for returns.

This ensures that all series share the same integration order, a precondition for Johansen cointegration analysis.

5.2 Johansen Cointegration

We apply the **Johansen (1988, 1991)** trace and maximum eigenvalue tests to determine the number of cointegrating relationships among the $N = 5$ exchange rates. The vector error correction representation is:

$$\Delta Y_t = \mu + \Pi Y_{t-1} + \sum_{i=1}^{p-1} \Gamma_i \Delta Y_{t-i} + \varepsilon_t, \quad (13)$$

where Y_t is the $N \times 1$ vector of log spot rates, $\Pi = \alpha\beta'$ captures long-run equilibrium relations, β contains the cointegrating vectors, and α the speed-of-adjustment coefficients.

Lag Length Selection. Lag order p is chosen using the **Akaike Information Criterion (AIC)**, **Schwarz Criterion (SC)**, and **Hannan–Quinn Criterion (HQ)**, supplemented by:

- **Sequential likelihood ratio (LR) tests**, and
- **VAR residual serial correlation LM tests**.

Lag selection is performed separately for each subperiod, as optimal lags can vary with volatility regimes.

Normalization in Johansen Tests. Johansen’s test is invariant to the choice of normalization; selecting a variable to normalize on only affects presentation of the cointegrating vector, not the rank. In our baseline, we normalize on the euro (EUR/USD) because:

1. It is the most liquid and least policy-managed non-USD currency.
2. It serves as a natural benchmark for global FX linkages.
3. Alternative numéraires (GBP, AUD, CHF, JPY) are more prone to idiosyncratic or regime-specific shocks.

Robustness checks with alternative normalizations yield qualitatively similar ranks.

- **Gregory–Hansen tests** for cointegration with structural breaks.
- **Regime-switching VECMs** to distinguish between temporary dislocations and permanent changes in equilibrium.

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Based on one single stochastic trend/factor driving all, but there are several

6 Complete Proofs of Propositions

Throughout this appendix, recall that

$$X_t = \mu_r + \Lambda_r g_t + \Gamma z_t + u_t,$$

with $\mu_r \in \mathbb{R}^N$, $\Lambda_r \in \mathbb{R}^N$, Γ is $N \times K$, z_t stationary and independent of ε_{t+1} , and u_t stationary; also $g_{t+1} - g_t = \mu_r + \sigma_{g,r} \varepsilon_{t+1}$.

Proof of Proposition 1 (Cointegration rank $N - 1$)

For any $b \in \mathbb{R}^N$, note:

$$b'X_t = b'\mu_r + (b'\Lambda_r)g_t + b'\Gamma z_t + b'u_t.$$

Since z_t and u_t are stationary, $b'\Gamma z_t + b'u_t$ is stationary. The only potentially nonstationary component is $(b'\Lambda_r)g_t$, due to g_t being $I(1)$:

- If $b'\Lambda_r = 0$, then $b'X_t$ is constant plus stationary variables $I(0)$. - If $b'\Lambda_r \neq 0$, then $b'X_t$ contains an $I(1)$ component $I(1)$.

Hence, the cointegrating space is $\{b : b'\Lambda_r = 0\}$, of dimension $N - 1$ when $\Lambda_r \neq 0$. Q.E.D.

Proof of Proposition 2 (Long-run Co-movement)

Write:

$$\Delta X_{i,t} = \lambda_{i,r} \Delta g_t + \Delta(\Gamma'_i z_t + u_{i,t}),$$

where the second term is stationary with finite variance not scaling with H . Let

$$Y_{i,H} = \sum_{h=1}^H \Delta X_{i,t+h} = \lambda_{i,r}(g_{t+H} - g_t) + \text{stationary term}.$$

Since

$$(g_{t+H} - g_t) = H \sigma_{g,r}^2,$$

we have:

$$\frac{1}{H}(Y_{i,H}) \rightarrow \lambda_{i,r}^2 \sigma_{g,r}^2, \quad \frac{1}{H}(Y_{i,H}, Y_{j,H}) \rightarrow \lambda_{i,r} \lambda_{j,r} \sigma_{g,r}^2,$$

so long-run correlation approaches $\text{sign}(\lambda_{i,r} \lambda_{j,r}) \in \{\pm 1\}$. Q.E.D.

Proof of Proposition 3 (Post-crisis Decoupling for EMs)

From $\lambda_{i,r} = (1 - \psi_{i,r}) \bar{\lambda}_i$. For $i \in \mathcal{E}$, if $\psi_{i,\text{post}} > \psi_{i,\text{crisis}}$, then:

$$|\lambda_{i,\text{post}}| = (1 - \psi_{i,\text{post}}) |\bar{\lambda}_i| < (1 - \psi_{i,\text{crisis}}) |\bar{\lambda}_i| = |\lambda_{i,\text{crisis}}|.$$

Given A1, $\sigma_{g,\text{post}}^2 \leq \sigma_{g,\text{crisis}}^2$, the global factor's contribution $\lambda_{i,r}^2$ strictly falls post-crisis for EMs; dispersion in those contributions increases. Q.E.D.

Proof of Proposition 4 (Error-Correction Half-Life)

Let $e_t = b'X_t$ with $b'\Lambda_r = 0$. The error-correction model is:

$$e_{t+1} = (1 - \kappa_r)e_t + \zeta_{t+1}.$$

Its solution:

$$e_{t+h} = (1 - \kappa_r)^h e_t + \sum_{j=1}^h (1 - \kappa_r)^{h-j} \zeta_{t+j}.$$

The half-life HL_r solves $(1 - \kappa_r)^{HL_r} = \frac{1}{2}$, so:

$$HL_r = \frac{\log(1/2)}{\log |1 - \kappa_r|},$$

which increases as κ_r decreases. Q.E.D.

Derivation of (7) from the SDF

Start from the key FX restriction:

$$1 = \mathbb{E}_t [M_{t+1} e^{X_{i,t} - X_{i,t+1}}].$$

Substitute $\log M_{t+1} = -r_t - \frac{1}{2}\zeta_r^2 - \lambda_r g_{t+1} - v_{t+1}$, express $X_{i,t+1}$ linearly in g_{t+1} plus stationary variables. Under joint normality, only a linear loading on g_t balances the term involving g_{t+1} . Solving yields:

$$X_{i,t} = \mu_{i,r} + \lambda_{i,r} g_t + \Gamma'_i z_t + u_{i,t},$$

with constants capturing Jensen corrections and stationary components as described. Q.E.D.

Deriving the affine form in (15). Start from the FX restriction

$$1 = \mathbb{E}_t [M_{t+1} e^{X_{i,t} - X_{i,t+1}}],$$

with $\log M_{t+1} = -r_t - \frac{1}{2}\zeta_r^2 - \xi_{t+1}$, $\xi_{t+1} = \lambda_r g_{t+1} + v_{t+1}$, $g_{t+1} = g_t + \mu_r + \sigma_{g,r}\varepsilon_{t+1}$, and $v_{t+1} \perp \varepsilon_{t+1}$, jointly normal. Conjecture (to be verified) that $X_{i,t} = A_{i,r} + B_{i,r}g_t + \Gamma'_i z_t + u_{i,t}$ with $z_t, u_{i,t}$ stationary and independent of ε_{t+1} . Then

$$\log(M_{t+1} e^{X_{i,t} - X_{i,t+1}}) = -r_t - \frac{1}{2}\zeta_r^2 + B_{i,r}g_t - (B_{i,r} + \lambda_r)g_{t+1} - v_{t+1} - \Gamma'_i(z_{t+1} - z_t) - (u_{i,t+1} - u_{i,t}).$$

Since the shocks are jointly normal and enter linearly, the conditional expectation is the exponential of the conditional mean plus one half the conditional variance. Writing $g_{t+1} = g_t + \mu_r + \sigma_{g,r}\varepsilon_{t+1}$, the deterministic (state-dependent) part contains $B_{i,r}g_t - (B_{i,r} + \lambda_r)(g_t + \mu_r) = -\lambda_r g_t - (B_{i,r} + \lambda_r)\mu_r$. For the restriction to hold for all states (g_t, z_t) , the coefficient on g_t must vanish. Hence $B_{i,r} = \lambda_{i,r}$. All remaining constants and stationary terms are absorbed into $\mu_{i,r}$, Γ_i , and $u_{i,t}$, yielding the affine solution $X_{i,t} = \mu_{i,r} + \lambda_{i,r}g_t + \Gamma'_i z_t + u_{i,t}$.

Proposition (Affine representation of $X_{i,t}$). Under regime r , suppose (i) the SDF is log-normal,

$$\log M_{t+1} = -r_t - \frac{1}{2}\zeta_r^2 - \xi_{t+1}, \quad \xi_{t+1} = \lambda_r g_{t+1} + v_{t+1},$$

with $\zeta_r^2 = \lambda_r^2 \sigma_{g,r}^2 + \sigma_v^2$; (ii) the global factor follows a random walk $g_{t+1} = g_t + \mu_r + \sigma_{g,r}\varepsilon_{t+1}$; (iii) ε_{t+1} and v_{t+1} are conditionally Gaussian and independent; (iv) the $K \times 1$ fundamentals z_t are stationary and independent of ε_{t+1} , and the stationary block $\Gamma'_i z_t + u_{i,t}$ has conditionally mean-zero first differences and finite variance.⁴ Then the no-arbitrage FX restriction

$$1 = \mathbb{E}_t [M_{t+1} e^{X_{i,t} - X_{i,t+1}}] \tag{14}$$

admits a log-linear (affine) solution

$$X_{i,t} = \mu_{i,r} + \lambda_{i,r}g_t + \Gamma'_i z_t + u_{i,t}, \quad u_{i,t} \text{ stationary}, \tag{15}$$

with $\lambda_{i,r}$ a currency- and regime-specific loading on the single nonstationary factor g_t , and

$$\mu_{i,r} = -\bar{r}_r - (\lambda_{i,r} + \lambda_r)\mu_r + \frac{1}{2}[(\lambda_{i,r} + \lambda_r)^2 \sigma_{g,r}^2 + \sigma_v^2 + (\Delta(\Gamma'_i z_t + u_{i,t}))], \tag{16}$$

⁴Any nonzero *constant* drift in the stationary block can be absorbed into the intercept below.

where $\bar{r}_r \equiv \mathbb{E}[r_t \mid r]$. Moreover, if $X_{i,t}$ had any additional I(1) component orthogonal to g_t , (14) would depend on that state and fail to hold for all $(g_t, \mathbf{z}_t)$; hence g_t spans the only nonstationary direction.

[Verification] Guess (15). Then

$$X_{i,t} - X_{i,t+1} = -\lambda_{i,r}(g_{t+1} - g_t) - \Delta(\Gamma'_i \mathbf{z}_t + u_{i,t}) = -\lambda_{i,r}\mu_r - \lambda_{i,r}\sigma_{g,r}\varepsilon_{t+1} - \Delta(\Gamma'_i \mathbf{z}_t + u_{i,t}).$$

Stack the innovations $\eta_{t+1} \equiv (\varepsilon_{t+1}, v_{t+1}, \Delta(\Gamma'_i \mathbf{z}_t + u_{i,t}))'$. By assumptions, $\eta_{t+1} \mid \mathcal{F}_t$ is (jointly) normal with mean zero and finite covariance, and $Y_{t+1} \equiv \log M_{t+1} + X_{i,t} - X_{i,t+1}$ is *linear* in η_{t+1} :

$$Y_{t+1} = -r_t - \frac{1}{2}\varsigma_r^2 - \lambda_r g_{t+1} + \lambda_{i,r} g_t - \lambda_{i,r}\mu_r - v_{t+1} - \lambda_{i,r}\sigma_{g,r}\varepsilon_{t+1} - \Delta(\Gamma'_i \mathbf{z}_t + u_{i,t}).$$

Using $g_{t+1} = g_t + \mu_r + \sigma_{g,r}\varepsilon_{t+1}$ gives

$$Y_{t+1} = \underbrace{-r_t - \frac{1}{2}\varsigma_r^2 - \lambda_r g_t - (\lambda_{i,r} + \lambda_r)\mu_r}_{\text{deterministic given } \mathcal{F}_t} + \underbrace{-(\lambda_{i,r} + \lambda_r)\sigma_{g,r}\varepsilon_{t+1} - v_{t+1} - \Delta(\Gamma'_i \mathbf{z}_t + u_{i,t})}_{\text{linear in Gaussian shocks}}.$$

Since $Y_{t+1} \mid \mathcal{F}_t$ is normal,

$$\mathbb{E}_t[e^{Y_{t+1}}] = \exp\left(\mathbb{E}_t[Y_{t+1}] + \frac{1}{2}_t(Y_{t+1})\right).$$

Imposing (14) therefore requires

$$\mathbb{E}_t[Y_{t+1}] + \frac{1}{2}_t(Y_{t+1}) = 0.$$

By the display above,

$$\mathbb{E}_t[Y_{t+1}] = -r_t - \frac{1}{2}\varsigma_r^2 - \lambda_r g_t - (\lambda_{i,r} + \lambda_r)\mu_r,$$

$$\frac{1}{2}_t(Y_{t+1}) = \frac{1}{2}[(\lambda_{i,r} + \lambda_r)^2\sigma_{g,r}^2 + \sigma_v^2 + \Delta(\Gamma'_i \mathbf{z}_t + u_{i,t})].$$

For $\mathbb{E}_t[Y_{t+1}] + \frac{1}{2}_t(Y_{t+1})$ to be *state-independent* (hold for all $(g_t, \mathbf{z}_t)$), any g_t -dependence in r_t must cancel the $-\lambda_r g_t$ term; this is the standard affine restriction on short rates.⁵ Taking expectations across r_t within the regime and collecting constants then yields (16). Finally, if $X_{i,t}$ contained any I(1) direction orthogonal to g_t , it would enter Y_{t+1} without a matching term in ξ_{t+1} , making $\mathbb{E}_t[e^{Y_{t+1}}]$ depend on that state and violating (14). Hence g_t is the *only*

⁵Allow the short rate to be affine in the state, $r_t = r_{0,r} + r_{g,r}g_t + r'_{z,r}\mathbf{z}_t$; the identity for all g_t implies $r_{g,r} = -\lambda_r$ and $r_{z,r} = 0$. This matches the “prices are exponential-affine in states” logic of affine pricing.

nonstationary driver, and (15) follows.

Start from the SDF SDE. Assume the stochastic discount factor M_t follows

$$\frac{dM_t}{M_t} = -r_t dt - \Lambda_t^\top dW_t, \quad (17)$$

where r_t is the short rate, W_t is a k -dimensional Brownian motion, and $\Lambda_t \in \mathbb{R}^k$ collects the (vector) prices of risk.

Itô's lemma for $\log M_t$. By Itô's formula,

$$d \log M_t = \frac{dM_t}{M_t} - \frac{1}{2} \left(\frac{dM_t}{M_t} \right)^2.$$

Using (17), note that $\left(\frac{dM_t}{M_t} \right)^2 = (\Lambda_t^\top dW_t)^2 = \|\Lambda_t\|^2 dt$. Hence

$$d \log M_t = -r_t dt - \Lambda_t^\top dW_t - \frac{1}{2} \|\Lambda_t\|^2 dt. \quad (18)$$

Integrate over a finite interval. Integrate (18) from t to $t + \Delta$:

$$\log M_{t+\Delta} - \log M_t = - \int_t^{t+\Delta} r_s ds - \frac{1}{2} \int_t^{t+\Delta} \|\Lambda_s\|^2 ds - \int_t^{t+\Delta} \Lambda_s^\top dW_s. \quad (19)$$

Define the (martingale) shock and its conditional variance:

$$\xi_{t+\Delta} \equiv \int_t^{t+\Delta} \Lambda_s^\top dW_s, \quad \varsigma^2 \equiv \int_t^{t+\Delta} \|\Lambda_s\|^2 ds.$$

Conditionally on $\mathcal{F}_t$, $\xi_{t+\Delta} \sim \mathcal{N}(0, \varsigma^2)$. Then (19) becomes

$$\log M_{t+\Delta} = \log M_t - \int_t^{t+\Delta} r_s ds - \frac{1}{2} \varsigma^2 - \xi_{t+\Delta}. \quad (20)$$

One-period, piecewise-constant specialization. If we take $\Delta = 1$ and assume $r_s \equiv r_t$ and $\Lambda_s \equiv \Lambda_t$ on $[t, t + 1]$, then

$$\int_t^{t+1} r_s ds = r_t, \quad \varsigma^2 = \int_t^{t+1} \|\Lambda_s\|^2 ds = \|\Lambda_t\|^2 \equiv \varsigma_r^2,$$

and $\xi_{t+1} = \Lambda_t^\top (W_{t+1} - W_t)$. Normalizing $M_t = 1$ (so $\log M_t = 0$), (20) reduces to

$$\boxed{\log M_{t+1} = -r_t - \frac{1}{2}\varsigma_r^2 - \xi_{t+1}}, \quad (21)$$

with $\xi_{t+1} | \mathcal{F}_t \sim \mathcal{N}(0, \varsigma_r^2)$.

Since $\xi_{t+1} \sim \mathcal{N}(0, \varsigma_r^2)$, $\mathbb{E}_t[e^{-\xi_{t+1}}] = \exp\left(\frac{1}{2}\varsigma_r^2\right)$. Therefore

$$\mathbb{E}_t[M_{t+1}] = \exp\left(-r_t - \frac{1}{2}\varsigma_r^2\right) \mathbb{E}_t[e^{-\xi_{t+1}}] = \exp(-r_t),$$

so $-\frac{1}{2}\varsigma_r^2$ is exactly the Jensen correction that pins $\mathbb{E}_t[M_{t+1}] = e^{-r_t}$.

Sketch of several

7 Complete Proofs of Propositions

Throughout this appendix, recall that in regime r the model solution is

$$X_t = \mu_r + \Lambda_r g_t + \Gamma z_t + u_t,$$

where:

- $\mu_r \in \mathbb{R}^N$ is a regime-specific intercept vector;
- $\Lambda_r \in \mathbb{R}^{N \times q_r}$ is the matrix of pass-through loadings on the q_r global nonstationary factors $g_t \in \mathbb{R}^{q_r}$;
- Γ is $N \times K$ and $z_t \in \mathbb{R}^K$ is stationary and independent of the global shocks ε_{t+1} ;
- u_t is stationary;
- The global factor vector follows $g_{t+1} - g_t = \mu_g + \Sigma_{g,r} \varepsilon_{t+1}$, with $\varepsilon_{t+1} \stackrel{i.i.d.}{\sim} \mathcal{N}(0, I_{q_r})$.

Proof of Proposition 1 (Cointegration rank $N - \text{rank}(\Lambda_r)$)

For any $b \in \mathbb{R}^N$,

$$b'X_t = b'\mu_r + (b'\Lambda_r)g_t + b'\Gamma z_t + b'u_t.$$

Since z_t and u_t are stationary, $b'\Gamma z_t + b'u_t$ is stationary. The only potentially nonstationary part is $(b'\Lambda_r)g_t$ because g_t is $I(1)$.

- If $b'\Lambda_r = 0_{1 \times q_r}$, then $b'X_t$ is constant plus stationary terms, hence $I(0)$.

- If $b'\Lambda_r \neq 0_{1 \times q_r}$, then $b'X_t$ contains at least one $I(1)$ component, hence $I(1)$.

Thus the cointegrating space is

$$\mathcal{C}_r = \{b \in \mathbb{R}^N : b'\Lambda_r = 0\},$$

whose dimension is $N - \text{rank}(\Lambda_r)$. In the special case $q_r = 1$ and $\Lambda_r \neq 0$, $\text{rank}(\Lambda_r) = 1$ and the cointegration rank is $N - 1$.

Proof of Proposition 2 (Alignment of Stochastic Trends)

Let $\lambda_{i,r}^\top$ denote the i th row of Λ_r (a $1 \times q_r$ vector of factor loadings). From the model,

$$\Delta X_{i,t} = \lambda_{i,r}^\top \Delta g_t + \Delta(\Gamma'_i z_t + u_{i,t}),$$

where the second term is stationary with variance not scaling with H . Let

$$Y_{i,H} = \sum_{h=1}^H \Delta X_{i,t+h} = \lambda_{i,r}^\top (g_{t+H} - g_t) + \text{stationary term}.$$

Since $\text{Var}(g_{t+H} - g_t) = H \Sigma_{g,r} \Sigma_{g,r}^\top$,

$$\frac{1}{H} \text{Var}(Y_{i,H}) \rightarrow \lambda_{i,r}^\top (\Sigma_{g,r} \Sigma_{g,r}^\top) \lambda_{i,r},$$

$$\frac{1}{H} \text{Cov}(Y_{i,H}, Y_{j,H}) \rightarrow \lambda_{i,r}^\top (\Sigma_{g,r} \Sigma_{g,r}^\top) \lambda_{j,r}.$$

The *alignment of stochastic trends* across i and j is determined by the angle between $\lambda_{i,r}$ and $\lambda_{j,r}$ in factor-loading space. If $\Sigma_{g,r} \Sigma_{g,r}^\top$ grows in magnitude (e.g., crisis volatility) and the $\lambda_{i,r}$ become more similar across i , the common-trend components in X_t become more aligned, increasing the persistence and magnitude of shared nonstationary variation.

Proof of Proposition 3 (Post-crisis Decoupling for EMs)

With g_t scalar ($q_r = 1$) for illustration, $\lambda_{i,r} = (1 - \psi_{i,r})\bar{\lambda}_i$. For $i \in \mathcal{E}$, if $\psi_{i,\text{post}} > \psi_{i,\text{crisis}}$, then

$$|\lambda_{i,\text{post}}| < |\lambda_{i,\text{crisis}}|.$$

Given A1, $\sigma_{g,\text{post}}^2 \leq \sigma_{g,\text{crisis}}^2$, the long-run variance contribution $\lambda_{i,r}^2 \sigma_{g,r}^2$ is strictly smaller post-crisis, increasing dispersion in exposures across currencies. For $q_r > 1$, the same holds

replacing $\lambda_{i,r}^2 \sigma_{g,r}^2$ with $\lambda_{i,r}^\top (\Sigma_{g,r} \Sigma_{g,r}^\top) \lambda_{i,r}$.

Proof of Proposition 4 (Error-Correction Half-Life)

Let $e_t = b' X_t$ with $b' \Lambda_r = 0$. Then e_t is stationary and follows the ECM

$$e_{t+1} = (1 - \kappa_r) e_t + \zeta_{t+1}.$$

Iterating forward,

$$e_{t+h} = (1 - \kappa_r)^h e_t + \sum_{j=1}^h (1 - \kappa_r)^{h-j} \zeta_{t+j}.$$

The half-life HL_r solves $(1 - \kappa_r)^{\text{HL}_r} = \frac{1}{2}$, giving

$$\text{HL}_r = \frac{\log(1/2)}{\log |1 - \kappa_r|},$$

which rises as κ_r falls.

Derivation of the affine form

From the no-arbitrage FX restriction.

$$1 = \mathbb{E}_t [M_{t+1} e^{X_{i,t} - X_{i,t+1}}],$$

with

$$\log M_{t+1} = -r_t - \frac{1}{2} \varsigma_r^2 - \lambda_r^\top g_{t+1} - v_{t+1}, \quad g_{t+1} = g_t + \mu_g + \Sigma_{g,r} \varepsilon_{t+1},$$

$v_{t+1} \perp \varepsilon_{t+1}$, jointly normal. Conjecture

$$X_{i,t} = \mu_{i,r} + \lambda_{i,r}^\top g_t + \Gamma_i' z_t + u_{i,t},$$

with stationary $z_t, u_{i,t}$ independent of ε_{t+1} . Substitute, separate deterministic and shock-driven parts, impose state-independence, and match g_t -coefficients. This forces $\lambda_{i,r}$ to be the row- i loading of Λ_r . Constants and stationary terms absorb into $\mu_{i,r}, \Gamma_i, u_{i,t}$, yielding the affine form and the intercept expression with Jensen corrections.

Proposition (Affine representation). [Include the detailed proof exactly as in your current draft, including (14)–(16), the Gaussian shock decomposition, and the footnote on affine short-rate restrictions.]

Start from the SDF SDE. [Retain the full Itô derivation (17)–(21), as in your original notes, showing the continuous-time setup and the discrete-time specialization, including the Jensen correction discussion.]

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